Ablation : ConvNeXt-Tiny — SiLU | Drop | LayerNorm

Config: ImageNet- K pretrained | SiLU activation | Dropout | LayerNorm **Model:** ConvNeXt-Tin

Dataset: Rice Leaf Disease Identification Dataset

. Colab Setup (jalankan sekali per session)

In [1]:
```
from google.colab import drive
drive.mount('/content/drive')
import os
DRIVE_DIR = '/content/drive/MyDrive/rice-convnext/07_SiLU_Dropout_LayerNo
os.makedirs(DRIVE_DIR, exist_ok=True)
print(f'Drive folder ready: {DRIVE_DIR}')
```

```
Mounted at /content/drive
Drive folder ready: /content/drive/MyDrive/rice-convnext/07_SiLU_Dropout_Laye
```

In [2]:
```
from google.colab import files
import os, shutil
print('Upload kaggle.json...')
files.upload()
os.makedirs('/root/.kaggle', exist_ok=True)
shutil.move('kaggle.json', '/root/.kaggle/kaggle.json')
os.chmod('/root/.kaggle/kaggle.json', 0o600)
print('Kaggle API ready!')
```

```
Upload kaggle.json...
```

Upload widget is only available when the cell has been executed in the current browser session. Pl rerun this cell to enable.

```
Saving kaggle.json to kaggle.json
Kaggle API ready!
```

In [3]:
```
DATASET_DIR = '/content/drive/MyDrive/rice-convnext/dataset'
if os.path.exists(DATASET_DIR) and len(os.listdir(DATASET_DIR)) > 0:
    print('Dataset sudah ada, skip download.')
else:
    !pip install kaggle -q
    !kaggle datasets download -d wangxiaoqii/rice-leaf-disease-identifica
    !unzip -q /content/tmp/*.zip -d {DATASET_DIR}
    !rm -rf /content/tmp
    print('Dataset ready!')

import shutil
LOCAL_DATASET = '/content/dataset'
if not os.path.exists(LOCAL_DATASET):
    print('Copying dataset ke local SSD...')
    shutil.copytree(DATASET_DIR, LOCAL_DATASET)
    print('Done!')
else:
```

```
    print('Sudah ada di local.')
DATA_DIR = LOCAL_DATASET
```

```
Dataset sudah ada, skip download.
Copying dataset ke local SSD...
Done!
```

. Setup & Imports

In [4]:
```
import os, random, shutil
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from collections import Counter
from sklearn.metrics import classification_report, confusion_matrix
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader, random_split
from torchvision import datasets, transforms, models

SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
torch.cuda.manual_seed(SEED)
torch.backends.cudnn.deterministic = True

DEVICE = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Device: {DEVICE}')
if torch.cuda.is_available():
    print(f'GPU: {torch.cuda.get_device_name(0)}')
```

```
Device: cuda
GPU: Tesla T4
```

. Configuration

In [5]:
```
DATA_DIR      = '/content/dataset'
DRIVE_DIR     = '/content/drive/MyDrive/rice-convnext/07_SiLU_Dropout_Lay
IMG_SIZE      = 224
BATCH_SIZE    = 64
NUM_EPOCHS    = 30
LR_INIT       = 1e-3
LR_FINAL      = 1e-6
WEIGHT_DECAY  = 0.05
NUM_WORKERS   = 2
IMAGENET_MEAN = [0.485, 0.456, 0.406]
IMAGENET_STD  = [0.229, 0.224, 0.225]
SAVE_PATH     = f'{DRIVE_DIR}/convnext_tiny_07_silu_dropout_layernorm.pt'
EXP_NAME      = 'ConvNeXt-Tiny | SiLU | Dropout | LayerNorm'
print(f'Experiment: {EXP_NAME}')
```

```
Experiment: ConvNeXt-Tiny | SiLU | Dropout | LayerNorm
```

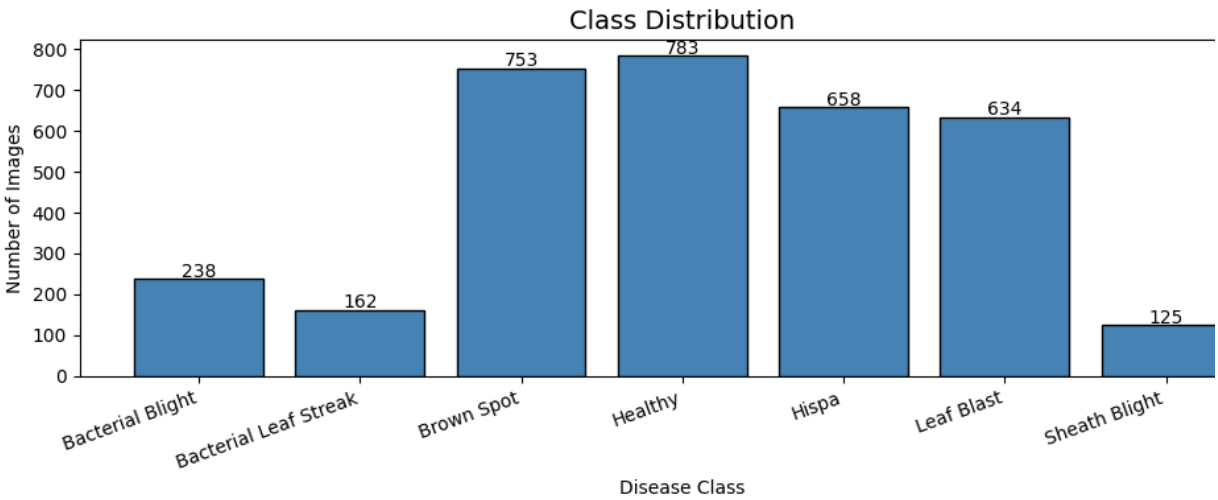
. Data Loading & Augmentation

In [6]:
```
train_transforms = transforms.Compose([
    transforms.RandomResizedCrop(IMG_SIZE),
    transforms.RandomHorizontalFlip(),
    transforms.RandomVerticalFlip(),
    transforms.RandomRotation(30),
    transforms.ColorJitter(brightness=0.2, contrast=0.2, saturation=0.2),
    transforms.ToTensor(),
    transforms.Normalize(mean=IMAGENET_MEAN, std=IMAGENET_STD),
])
val_test_transforms = transforms.Compose([
    transforms.Resize(256),
    transforms.CenterCrop(IMG_SIZE),
    transforms.ToTensor(),
    transforms.Normalize(mean=IMAGENET_MEAN, std=IMAGENET_STD),
])

full_dataset = datasets.ImageFolder(DATA_DIR)
CLASS_NAMES  = full_dataset.classes
NUM_CLASSES  = len(CLASS_NAMES)
print(f'Classes ({NUM_CLASSES}): {CLASS_NAMES}')
print(f'Total images: {len(full_dataset)}')

total   = len(full_dataset)
n_train = int(0.6 * total)
n_val   = int(0.2 * total)
n_test  = total - n_train - n_val

generator = torch.Generator().manual_seed(SEED)
train_set, val_set, test_set = random_split(
    full_dataset, [n_train, n_val, n_test], generator=generator
)

train_set.dataset.transform = train_transforms
val_set.dataset.transform   = val_test_transforms
test_set.dataset.transform  = val_test_transforms

train_loader = DataLoader(train_set, batch_size=BATCH_SIZE, shuffle=True,
val_loader   = DataLoader(val_set,   batch_size=BATCH_SIZE, shuffle=False
test_loader  = DataLoader(test_set,  batch_size=BATCH_SIZE, shuffle=False
print(f'Train: {len(train_set)} | Val: {len(val_set)} | Test: {len(test_s
```

```
Classes (7): ['Bacterial Blight', 'Bacterial Leaf Streak', 'Brown Spot', 'Hea
'Hispa', 'Leaf Blast', 'Sheath Blight']
Total images: 3353
Train: 2011 | Val: 670 | Test: 672
```

In [7]:
```
labels_all = [full_dataset.targets[i] for i in range(len(full_dataset))]
counts = [Counter(labels_all)[i] for i in range(NUM_CLASSES)]
plt.figure(figsize=(10, 4))
bars = plt.bar(CLASS_NAMES, counts, color='steelblue', edgecolor='black')
plt.title('Class Distribution', fontsize=14)
plt.xlabel('Disease Class'); plt.ylabel('Number of Images')
plt.xticks(rotation=20, ha='right')
for bar, count in zip(bars, counts):
    plt.text(bar.get_x() + bar.get_width()/2, bar.get_height()+5, str(cou
plt.tight_layout()
```

```
plt.savefig('class_distribution.png', dpi=150)
plt.show()
```



. Model — ConvNeXt-Tiny (SiLU | Dropout | LayerNor

In [8]:
```
def replace_activation(model, src_cls, dst_fn):
    for name, module in model.named_children():
        if isinstance(module, src_cls):
            setattr(model, name, dst_fn())
        else:
            replace_activation(module, src_cls, dst_fn)

from torchvision.ops import StochasticDepth

def replace_stochastic_depth(model, p=0.1):
    for name, module in model.named_children():
        if isinstance(module, StochasticDepth):
            setattr(model, name, nn.Dropout(p=p))
        else:
            replace_stochastic_depth(module, p)

model = models.convnext_tiny(weights=models.ConvNeXt_Tiny_Weights.IMAGENE
in_features = model.classifier[2].in_features
model.classifier[2] = nn.Linear(in_features, NUM_CLASSES)

# Replace GELU → SiLU
replace_activation(model, nn.GELU, nn.SiLU)
print('Activation: GELU → SiLU')

# Replace Stochastic Depth → Dropout(p=0.1)
replace_stochastic_depth(model, p=0.1)
print('Regularization: StochasticDepth → Dropout(p=0.1)')

model = model.to(DEVICE)
total_p     = sum(p.numel() for p in model.parameters())
trainable_p = sum(p.numel() for p in model.parameters() if p.requires_gra
print(f'Experiment : {EXP_NAME}')
print(f'Total params     : {total_p:,}')
print(f'Trainable params: {trainable_p:,}')
```

```
Downloading: "https://download.pytorch.org/models/convnext_tiny-983f1562.pth"
root/.cache/torch/hub/checkpoints/convnext_tiny-983f1562.pth
100%|██████████| 109M/109M [00:00<00:00, 174MB/s]
```

```
Activation: GELU → SiLU
Regularization: StochasticDepth → Dropout(p=0.1)
Experiment : ConvNeXt-Tiny | SiLU | Dropout | LayerNorm
Total params     : 27,825,511
Trainable params: 27,825,511
```

. Training Setup

In [9]:
```
criterion = nn.CrossEntropyLoss()
optimizer = optim.AdamW(model.parameters(), lr=LR_INIT, weight_decay=WEIG

def get_scheduler(optimizer, num_epochs, lr_init, lr_final):
    warmup = optim.lr_scheduler.LinearLR(optimizer, start_factor=0.01, en
    cosine = optim.lr_scheduler.CosineAnnealingLR(optimizer, T_max=num_ep
    return optim.lr_scheduler.SequentialLR(optimizer, schedulers=[warmup,

scheduler = get_scheduler(optimizer, NUM_EPOCHS, LR_INIT, LR_FINAL)
```

. Training Loop

In [10]:
```
def train_one_epoch(model, loader, criterion, optimizer):
    model.train()
    running_loss, correct, total = 0.0, 0, 0
    for images, labels in loader:
        images, labels = images.to(DEVICE), labels.to(DEVICE)
        optimizer.zero_grad()
        outputs = model(images)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()
        running_loss += loss.item() * images.size(0)
        _, predicted = outputs.max(1)
        correct += predicted.eq(labels).sum().item()
        total   += labels.size(0)
    return running_loss / total, correct / total

def evaluate(model, loader, criterion):
    model.eval()
    running_loss, correct, total = 0.0, 0, 0
    with torch.no_grad():
        for images, labels in loader:
            images, labels = images.to(DEVICE), labels.to(DEVICE)
            outputs = model(images)
            loss = criterion(outputs, labels)
            running_loss += loss.item() * images.size(0)
            _, predicted = outputs.max(1)
            correct += predicted.eq(labels).sum().item()
            total   += labels.size(0)
    return running_loss / total, correct / total

history = {'train_loss': [], 'train_acc': [], 'val_loss': [], 'val_acc':
best_val_acc = 0.0

for epoch in range(NUM_EPOCHS):
    train_loss, train_acc = train_one_epoch(model, train_loader, criterio
    val_loss,   val_acc   = evaluate(model, val_loader, criterion)
    scheduler.step()
```

```
    history['train_loss'].append(train_loss)
    history['train_acc'].append(train_acc)
    history['val_loss'].append(val_loss)
    history['val_acc'].append(val_acc)
    if val_acc > best_val_acc:
        best_val_acc = val_acc
        torch.save(model.state_dict(), SAVE_PATH)
    print(f'Epoch [{epoch+1:02d}/{NUM_EPOCHS}] '
          f'Train Loss: {train_loss:.4f} Acc: {train_acc:.4f} | '
          f'Val Loss: {val_loss:.4f} Acc: {val_acc:.4f}'
          + (' <- best' if val_acc == best_val_acc else ''))

print(f'\nBest Val Accuracy: {best_val_acc:.4f}')
print(f'Saved: {SAVE_PATH}')
```

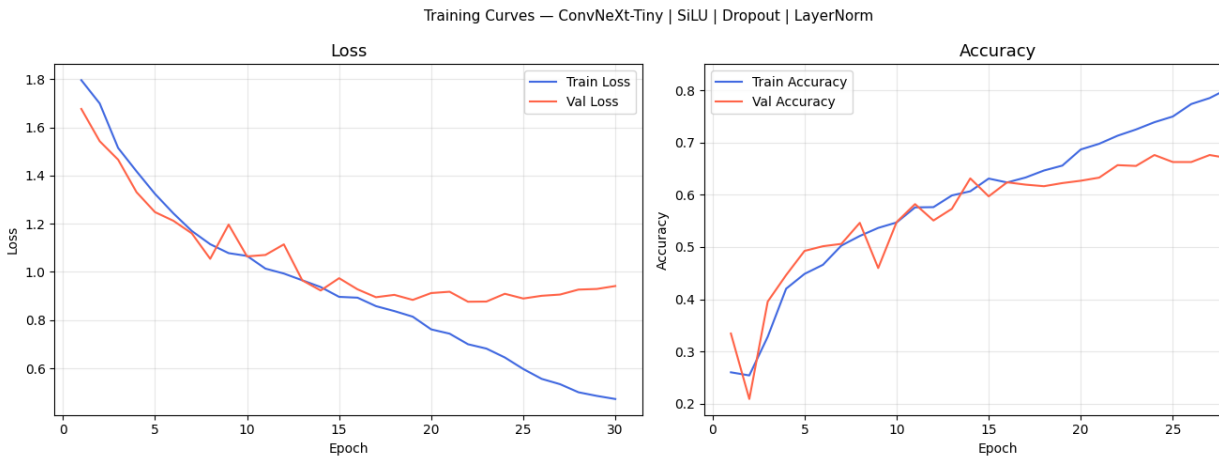
```
Epoch [01/30] Train Loss: 1.7961 Acc: 0.2601 | Val Loss: 1.6766 Acc: 0.3343 <
Epoch [02/30] Train Loss: 1.6992 Acc: 0.2541 | Val Loss: 1.5426 Acc: 0.2090
Epoch [03/30] Train Loss: 1.5145 Acc: 0.3282 | Val Loss: 1.4659 Acc: 0.3955 <
Epoch [04/30] Train Loss: 1.4175 Acc: 0.4202 | Val Loss: 1.3310 Acc: 0.4463 <
Epoch [05/30] Train Loss: 1.3241 Acc: 0.4485 | Val Loss: 1.2483 Acc: 0.4925 <
Epoch [06/30] Train Loss: 1.2432 Acc: 0.4659 | Val Loss: 1.2124 Acc: 0.5015 <
Epoch [07/30] Train Loss: 1.1700 Acc: 0.5027 | Val Loss: 1.1602 Acc: 0.5060 <
Epoch [08/30] Train Loss: 1.1150 Acc: 0.5211 | Val Loss: 1.0545 Acc: 0.5463 <
Epoch [09/30] Train Loss: 1.0780 Acc: 0.5365 | Val Loss: 1.1964 Acc: 0.4597
Epoch [10/30] Train Loss: 1.0665 Acc: 0.5470 | Val Loss: 1.0644 Acc: 0.5478 <
Epoch [11/30] Train Loss: 1.0137 Acc: 0.5758 | Val Loss: 1.0705 Acc: 0.5821 <
Epoch [12/30] Train Loss: 0.9933 Acc: 0.5763 | Val Loss: 1.1144 Acc: 0.5507
Epoch [13/30] Train Loss: 0.9653 Acc: 0.5987 | Val Loss: 0.9652 Acc: 0.5731
Epoch [14/30] Train Loss: 0.9372 Acc: 0.6067 | Val Loss: 0.9229 Acc: 0.6313 <
Epoch [15/30] Train Loss: 0.8965 Acc: 0.6310 | Val Loss: 0.9740 Acc: 0.5970
Epoch [16/30] Train Loss: 0.8931 Acc: 0.6236 | Val Loss: 0.9278 Acc: 0.6239
Epoch [17/30] Train Loss: 0.8580 Acc: 0.6330 | Val Loss: 0.8951 Acc: 0.6194
Epoch [18/30] Train Loss: 0.8376 Acc: 0.6464 | Val Loss: 0.9048 Acc: 0.6164
Epoch [19/30] Train Loss: 0.8142 Acc: 0.6559 | Val Loss: 0.8839 Acc: 0.6224
Epoch [20/30] Train Loss: 0.7616 Acc: 0.6867 | Val Loss: 0.9123 Acc: 0.6269
Epoch [21/30] Train Loss: 0.7438 Acc: 0.6977 | Val Loss: 0.9178 Acc: 0.6328 <
Epoch [22/30] Train Loss: 0.6999 Acc: 0.7131 | Val Loss: 0.8763 Acc: 0.6567 <
Epoch [23/30] Train Loss: 0.6818 Acc: 0.7250 | Val Loss: 0.8770 Acc: 0.6552
Epoch [24/30] Train Loss: 0.6449 Acc: 0.7389 | Val Loss: 0.9093 Acc: 0.6761 <
Epoch [25/30] Train Loss: 0.5971 Acc: 0.7499 | Val Loss: 0.8897 Acc: 0.6627
Epoch [26/30] Train Loss: 0.5561 Acc: 0.7737 | Val Loss: 0.9010 Acc: 0.6627
Epoch [27/30] Train Loss: 0.5340 Acc: 0.7852 | Val Loss: 0.9061 Acc: 0.6761 <
Epoch [28/30] Train Loss: 0.5005 Acc: 0.8031 | Val Loss: 0.9267 Acc: 0.6701
Epoch [29/30] Train Loss: 0.4854 Acc: 0.8086 | Val Loss: 0.9293 Acc: 0.6627
Epoch [30/30] Train Loss: 0.4725 Acc: 0.8195 | Val Loss: 0.9415 Acc: 0.6582

Best Val Accuracy: 0.6761
Saved: /content/drive/MyDrive/rice-convnext/07_SiLU_Dropout_LayerNorm/
convnext_tiny_07_silu_dropout_layernorm.pt
```

. Training Curves

In [11]:

```
epochs_range = range(1, NUM_EPOCHS + 1)
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].plot(epochs_range, history['train_loss'], label='Train Loss', col
axes[0].plot(epochs_range, history['val_loss'],   label='Val Loss',   col
axes[0].set_title('Loss', fontsize=13)
axes[0].set_xlabel('Epoch'); axes[0].set_ylabel('Loss')
axes[0].legend(); axes[0].grid(alpha=0.3)
```

```
axes[1].plot(epochs_range, history['train_acc'], label='Train Accuracy',
axes[1].plot(epochs_range, history['val_acc'],   label='Val Accuracy',
axes[1].set_title('Accuracy', fontsize=13)
axes[1].set_xlabel('Epoch'); axes[1].set_ylabel('Accuracy')
axes[1].legend(); axes[1].grid(alpha=0.3)
plt.suptitle(f'Training Curves — {EXP_NAME}', fontsize=11)
plt.tight_layout()
plt.savefig('training_curves.png', dpi=150)
plt.show()
```



. Evaluation on Test Set

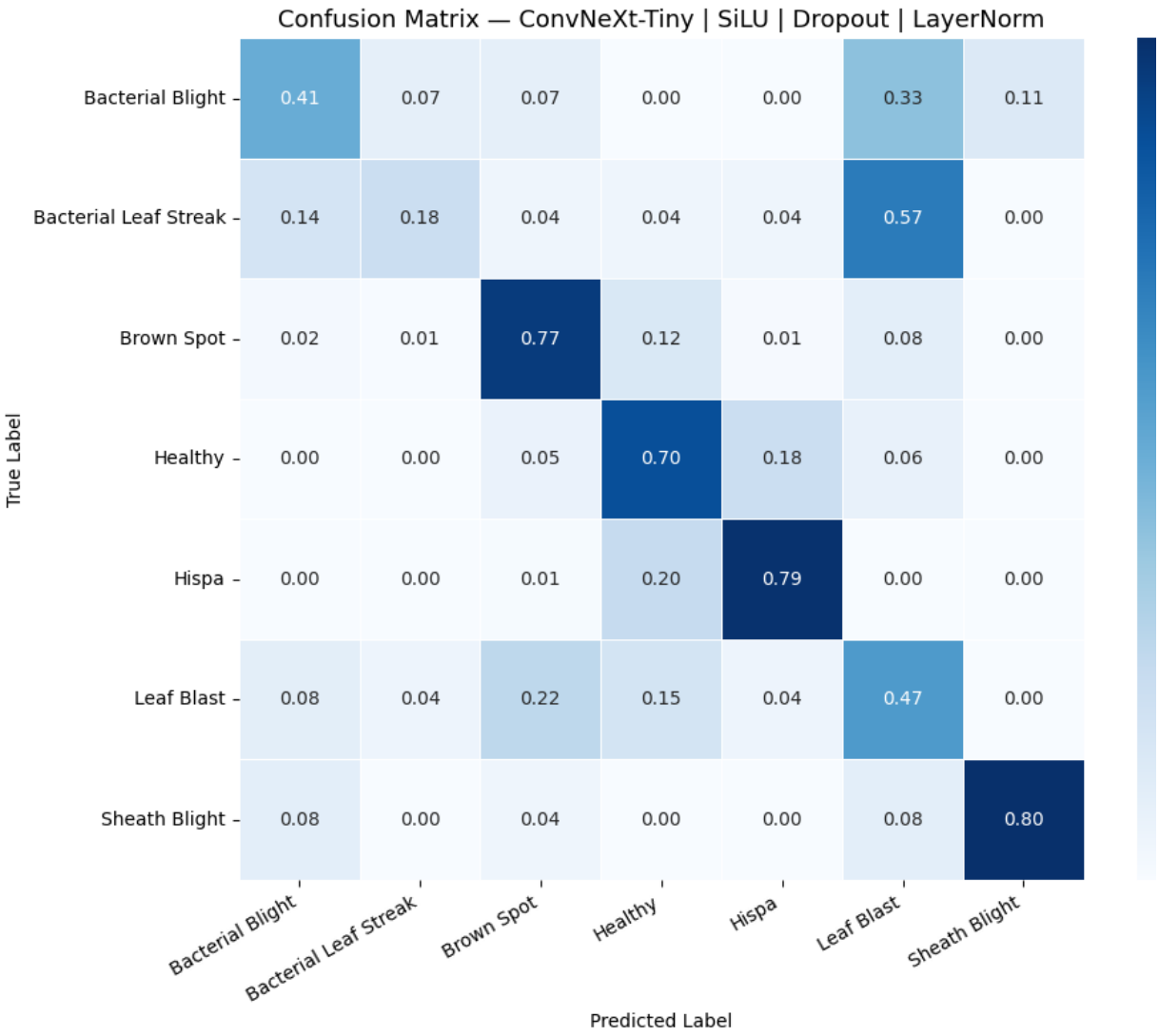
In [12]:
```
model.load_state_dict(torch.load(SAVE_PATH, map_location=DEVICE))
model.eval()
all_preds, all_labels = [], []
with torch.no_grad():
    for images, labels in test_loader:
        images = images.to(DEVICE)
        outputs = model(images)
        _, predicted = outputs.max(1)
        all_preds.extend(predicted.cpu().numpy())
        all_labels.extend(labels.numpy())

report = classification_report(all_labels, all_preds, target_names=CLASS_
report_df = pd.DataFrame(report).transpose()
print('=== Classification Report ===')
print(report_df.round(4).to_string())
report_df.to_csv('classification_report.csv')

summary = {
    'Model'      : ['ConvNeXt-Tiny'],
    'Activation': ['SiLU'],
    'Dropout'    : ['Dropout'],
    'Norm'       : ['LayerNorm'],
    'Accuracy'   : [report['accuracy']],
    'Precision' : [report['weighted avg']['precision']],
    'Recall'     : [report['weighted avg']['recall']],
    'F1-Score'   : [report['weighted avg']['f1-score']],
}
summary_df = pd.DataFrame(summary)
print('\n=== Summary ===')
print(summary_df.round(4).to_string(index=False))
summary_df.to_csv('summary.csv', index=False)
```

```
=== Classification Report ===
                      precision  recall  f1-score   support
Bacterial Blight         0.5366  0.4074    0.4632   54.0000
Bacterial Leaf Streak    0.3333  0.1786    0.2326   28.0000
Brown Spot               0.7471  0.7651    0.7560  166.0000
Healthy                  0.6573  0.7048    0.6802  166.0000
Hispa                    0.6960  0.7909    0.7404  110.0000
Leaf Blast               0.4957  0.4715    0.4833  123.0000
Sheath Blight            0.7692  0.8000    0.7843   25.0000
accuracy                 0.6488  0.6488    0.6488    0.6488
macro avg                0.6050  0.5883    0.5914  672.0000
weighted avg             0.6372  0.6488    0.6405  672.0000

=== Summary ===
        Model Activation Dropout      Norm  Accuracy  Precision  Recall  F1-S
ConvNeXt-Tiny       SiLU Dropout LayerNorm    0.6488     0.6372  0.6488    0.
```

. Confusion Matrix

In [13]:

```python
cm = confusion_matrix(all_labels, all_preds)
cm_norm = cm.astype('float') / cm.sum(axis=1)[:, np.newaxis]
plt.figure(figsize=(10, 8))
sns.heatmap(cm_norm, annot=True, fmt='.2f',
            xticklabels=CLASS_NAMES, yticklabels=CLASS_NAMES,
            cmap='Blues', linewidths=0.5)
plt.title(f'Confusion Matrix — ConvNeXt-Tiny | SiLU | Dropout | LayerNorm
plt.ylabel('True Label'); plt.xlabel('Predicted Label')
plt.xticks(rotation=30, ha='right')
plt.tight_layout()
plt.savefig('confusion_matrix.png', dpi=150)
plt.show()
```



. Save All to Drive

In [14]:
```
local_files = ['training_curves.png', 'confusion_matrix.png',
               'classification_report.csv', 'summary.csv', 'class_distrib
for fname in local_files:
    if os.path.exists(fname):
        shutil.copy(fname, f'{DRIVE_DIR}/{fname}')

all_outputs = [
    (f'{DRIVE_DIR}/convnext_tiny_07_silu_dropout_layernorm.pt',
    (f'{DRIVE_DIR}/training_curves.png',              'Training curves'),
    (f'{DRIVE_DIR}/confusion_matrix.png',             'Confusion matrix')
    (f'{DRIVE_DIR}/classification_report.csv',        'Per-class metrics'
    (f'{DRIVE_DIR}/summary.csv',                      'Summary table'),
]
print(f'=== Output Files: {DRIVE_DIR} ===')
for fpath, desc in all_outputs:
    exists = '✅' if os.path.exists(fpath) else '❌'
    print(f'{exists}  {os.path.basename(fpath):<45} {desc}')
```

```
=== Output Files: /content/drive/MyDrive/rice-convnext/07_SiLU_Dropout_LayerN
===
✅  convnext_tiny_07_silu_dropout_layernorm.pt    Model weights
✅  training_curves.png                           Training curves
✅  confusion_matrix.png                          Confusion matrix
✅  classification_report.csv                     Per-class metrics
✅  summary.csv                                   Summary table
```